# AI453 — Practical #2: Conditional Independence

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**Not assessed — nothing is collected.** Plain Python 3, nothing to install. Run `lab2_independence.py`; it prints `P(A=1) = 0.5`. The rest is yours.

**Table 1** — $P(A, B, C)$

| $A$ | $B$ | $C$ | $P$ |
|---|---|---|---|
| 0 | 0 | 0 | 0.36 |
| 0 | 0 | 1 | 0.04 |
| 0 | 1 | 0 | 0.01 |
| 0 | 1 | 1 | 0.09 |
| 1 | 0 | 0 | 0.09 |
| 1 | 0 | 1 | 0.01 |
| 1 | 1 | 0 | 0.04 |
| 1 | 1 | 1 | 0.36 |

**Table 2** — $P(R, S, W)$

| $R$ | $S$ | $W$ | $P$ |
|---|---|---|---|
| 0 | 0 | 0 | 0.27 |
| 0 | 0 | 1 | 0.03 |
| 0 | 1 | 0 | 0.12 |
| 0 | 1 | 1 | 0.18 |
| 1 | 0 | 0 | 0.08 |
| 1 | 0 | 1 | 0.12 |
| 1 | 1 | 0 | 0.02 |
| 1 | 1 | 1 | 0.18 |

$R$ = it rained, $S$ = sprinkler was on, $W$ = grass is wet. Table 1 has no story.

Two ready-made functions in the file — last week's loop, wrapped up:

```
prob(P1, {0: 1})             P(A=1)              position 0=A, 1=B, 2=C
prob(P1, {0: 1, 2: 1})       P(A=1, C=1)
cond(P1, {2: 1}, {0: 1})     P(C=1 | A=1)        query first, then given
```

## Tasks

### Table 1

**T1.** Print the total of each table. Both should be 1.

**T2.** Print $P(A=1, C=1)$ and $P(A=1)\,P(C=1)$. Equal? If not, $A$ and $C$ are dependent.

**T3.** Print $P(C=1 \mid A=1)$ and $P(C=1 \mid A=0)$. How far apart?

**T4.** Print $P(C=1 \mid B=1)$, $P(C=1 \mid B=1, A=1)$, $P(C=1 \mid B=1, A=0)$.
*Once $B$ is known, does $A$ still change anything?*

**T5.** Same three for $B=0$. Then finish this line in a comment:
*"$A$ and $C$ are __________, but __________ given $B$."*

### Table 2

**T6.** Print $P(R=1, S=1)$ and $P(R=1)\,P(S=1)$. These should agree — rain and sprinklers are unrelated.

**T7.** Print $P(R=1)$, $P(R=1 \mid W=1)$, $P(R=1 \mid W=1, S=1)$, $P(R=1 \mid W=1, S=0)$.

**T8.** Two or three lines in a comment. In T5 conditioning **removed** a dependence; in T7 it **created** one. Why does learning the sprinkler was on make rain *less* likely, when the grass is just as wet either way?

**Do T1–T3 and T6 on paper first**

Each is a handful of numbers — ten seconds with a pen. Then check the computer agrees. T4, T5 and T7 are worth running.